Name	Ben Chong Wei Heng
Student ID	34860398
Studio Session	Studio 04 Friday 10.00 AM – 12.00 PM

Part A: URL of the Visualisation

Paste your visualisation URL in the empty space below.
Tip: Test your URL in incognito mode to ensure it is a working URL

https://benc004.github.io/3179/Project%202/

Part B: Domain Description

Describe your domain chosen for DV II
My domain for DV2 is School Enrolment Trends in Malaysia. This domain focuses on government primary and secondary education from 2017 – 2025. The visualization explores enrolment differences across states and school stages. This domain also explores how the trend change with student gender composition and demographic changes.

Describe the Why and Who aspects of your DV II
The purpose of this data visualization is to help audiences understand school enrolment trends in Malaysia. It explains how a relatively stable national total hides differences across state and school stages. It allows audience to understand where enrolment increases and decreases. They can also compare these changes in births and the child population. Furthermore, it also shows school distribution and teacher provision.

The visualization is designed for average Malaysian, especially parents, educators, students, and people who are interested in education.

Describe the "What" aspect of your DV II
This visualization uses two real-world data sources. The first source is the Ministry of Education (MOE), accessed through data.gov.my. It provides data on student enrolment, school counts and teacher counts. These datasets contain national, state and district-level records. This visualization focuses on government primary and secondary education from 2017 to 2025. This dataset is relevant as it provides information that helps identify enrolment differences.

(Student Enrolment, https://data.gov.my/data-catalogue/enrolment_school_district)
(School Counts, https://data.gov.my/data-catalogue/schools_district)
(Teacher Counts, https://data.gov.my/data-catalogue/teachers_district)

The second source is population data accessed through OpenDOSM. The population state dataset provides age, sex, and ethnicity data. The visualization compares population numbers from 2017 and 2025. It also uses data with annual registered births from 2017 to 2024. These datasets provide information on enrolment and future school demands.
(Population state, https://open.dosm.gov.my/data-catalogue/population_state)
(Annual birth state, https://open.dosm.gov.my/data-catalogue/births_annual_state)

The data are stored as CSV files for Vega and Vega-Lite. Relevant records are selected, grouped, and combined for the chart. A GeoJSON file provides the state boundaries for the maps.

Describe the "How" aspect of your DV II

The visualization is built using Vega and Vega-Lite, with HTML, CSS, and JavaScript. It guides readers from national enrolment to state differences. It then introduces population changes, schools, and teacher provision. Each chart answers a specific question within this story.

1. **Multi-line chart**

A multi-line chart was used to compare primary and secondary enrolment from 2017 to 2025. These lines help audiences follow changed over time and compare both school stages.
Marks: Line and Point
Channels:

- Position shows years and student numbers.
- Colour hue represents primary and secondary stages.

2. **Waterfall chart**

A waterfall chart was used to show each state's contribution to the national enrolment change. It helps audiences understand how increases and decreases balance out each other.
Marks: Line representing bars and the reference line
Channels:

- Length shows each state's enrolment change.
- Position shows cumulative totals.
- Colour hue represents increases and decreases.
- Horizontal spatial region separates the states.

3. Choropleth map

A choropleth map was used to compare enrolment percentage changes across states. The colours help audiences identify geographic patterns of growth and decline. Colour scale is capped at +- 20%.

Marks: Area

Channels:

- Spatial region identifies each state.
- Colour hue represents increases and decreases.
- Colour luminance shows the magnitude within each colour distribution.

4. Proportional symbol map

A proportional symbol map was used to show student concentration across states. It appears beside the choropleth to compare student numbers with percentage growth. Larger circles indicate more students.

Marks: Point

Channels:

- Area represents student numbers through circle size.

5. Heatmap

A heatmap was used to show annual enrolment changes across states. It helps audiences identify patterns, unusual changes, and differences between each year. Colour scale is capped at +- 5%.

Marks: Area

Channels:

- Colour hue represents increases and decreases.
- Colour luminance shows the magnitude within the displayed range.

6. Waffle chart

A waffle chart was used to compare boys' and girls' enrolment distribution. The grid makes proportions easier for audience to understand and compare across school stages. Each full square represents one percentage point.

Marks: Area

Channels:

- Area represents each group's distributions.
- Colour hue represents boys and girls.

7. Parallel coordinates plot

A parallel coordinates plot was used to compare child population changes with primary and secondary enrolment changes. Connected lines help audiences follow each state across three measures.

Marks: Line and Point

Channels:

- Spatial region separates the three measure axes.
- Vertical position shows percentage changes.

8. Paired population pyramid

A paired population pyramid was used to compare the young population in 2017 and 2025. It helps audiences identify which age groups became larger or smaller.

Marks: Line

Channels:

- Length represents population count.
- Vertical position orders age groups
- Colour hue represents males and females.

9. Horizon graph

A horizon graph was used to compare birth trends across locations. Each location uses its own 2017 birth count as its starting point.

Marks: Area

Channels:

- Colour hue represents increases and decreases.
- Colour luminance shows the magnitude within the displayed range.
- Height (length) to show the remaining change within the displayed range.

10. Non-contiguous cartogram

A non-contiguous cartogram was used to compare school distribution. It helps audiences distinguish school numbers from average school sizes. Size and colour represent different measures.

Marks: Area represents resized state outlines

Channels:

- Area represents school count.
- Colour luminance represents students per school.

11. Scatterplot

A scatterplot was used to compare teacher growth with enrolment growth. The diagonal reference line helps audiences identify which measure grew faster. Purple dots represent primary education while orange square represent secondary education.

Marks: Point

Channels:

- Horizontal position shows enrolment growth.
- Vertical position shows teacher growth.
- Colour hue represents primary and secondary stages.
- Shape represents primary and secondary stages.

12. Boxplot

A boxplot was used to compare district students-per-teacher ratios within each state. They reveal differences hidden within state averages.

Marks: Area for boxes, Line for medians and whiskers and point for districts

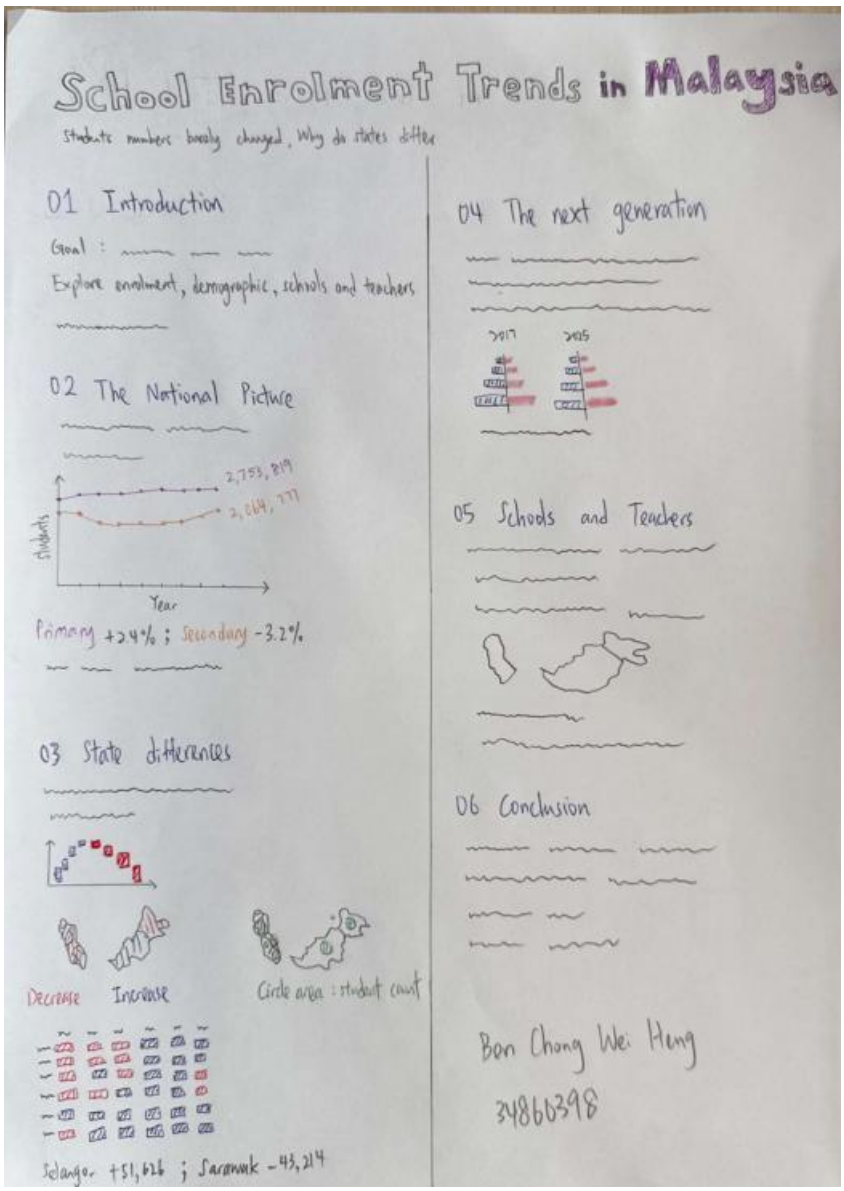
Channels:

- Horizontal position represents staffing ratios.
- Length shows the middle-half spread through box width.
- Colour hue represents primary and secondary stages.

Part C: Sketch

A URL to a PDF document with an image of your sketch on your GitHub account.
Tip: Test your URL in incognito mode to ensure it is a working URL

https://benc004.github.io/3179/Project%202/DV2_Sketch.pdf

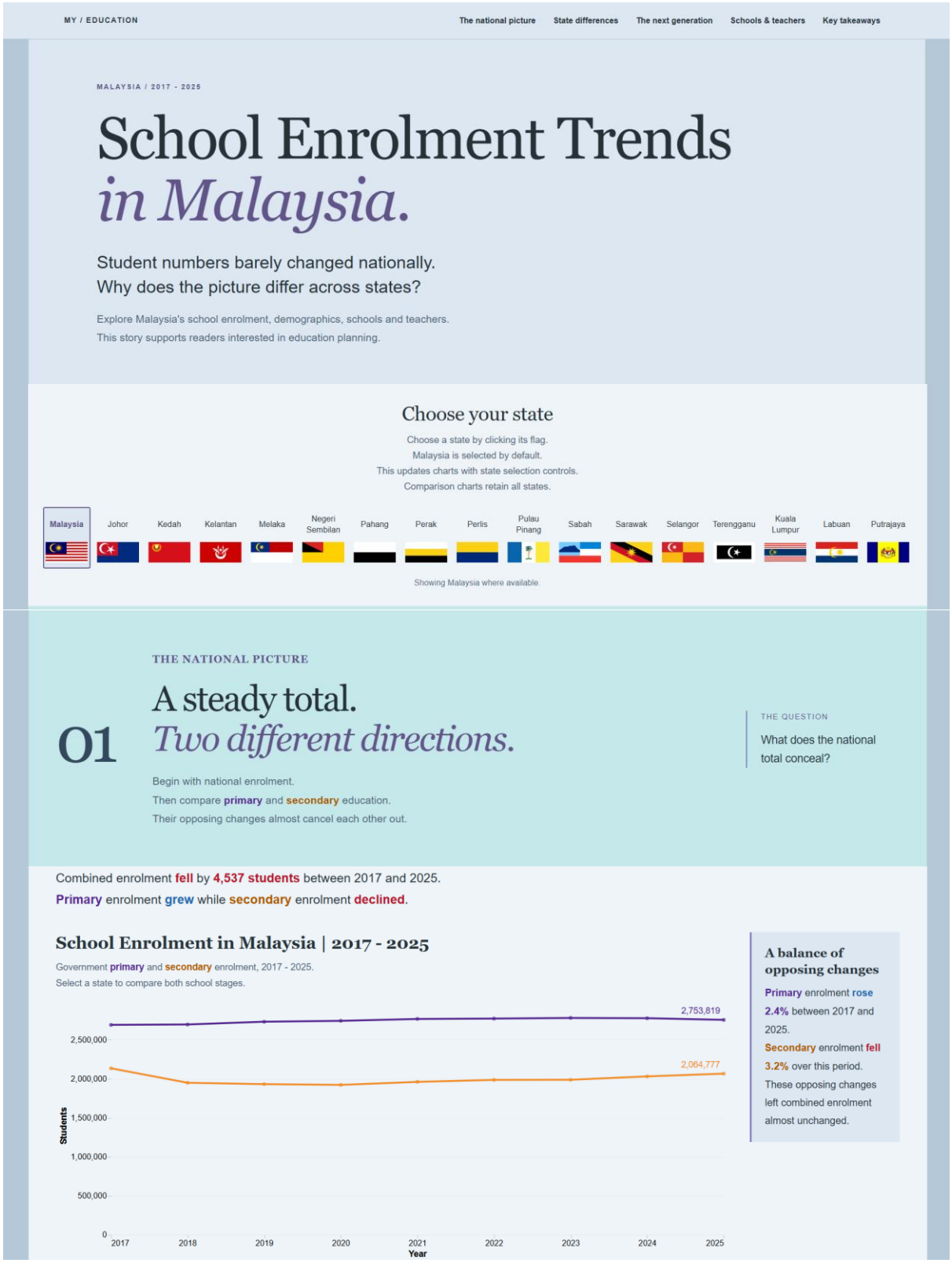


Paste a screenshot of your sketch for your DV II.
Note: Use the entire page here for the screenshot and ensure that it is clearly legible and visible

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Part D: Full-size screenshot of your DV II

Paste a full-size screenshot of your DV II assignment from the URL here



School stage
● Primary ● Secondary

Selected state: Malaysia

WHAT THIS TELLS US The national total hides differences between school stages.

STATE DIFFERENCES

02 One country. *Uneven change.*

The same balancing effect appears across states.
Compare student **gains** and **losses** first.
Then distinguish percentage growth from additional student numbers.

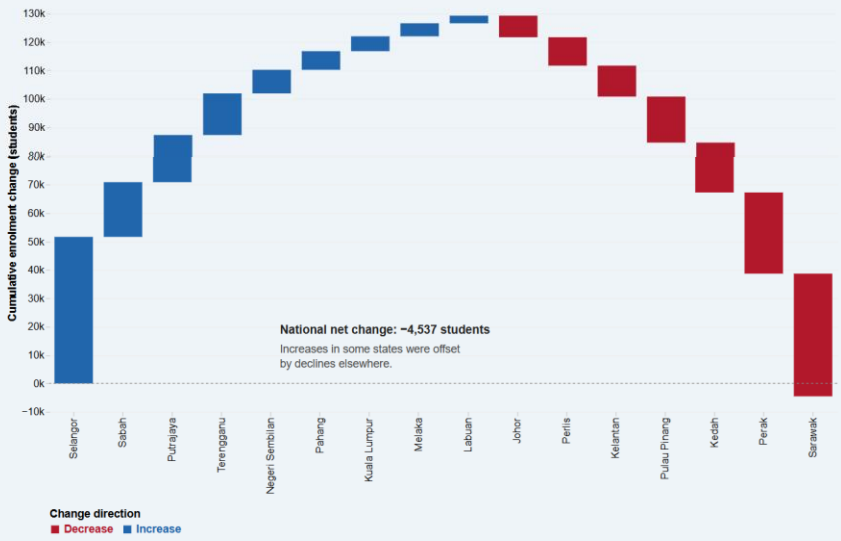
THE QUESTION

Where is enrolment growing, and where is it declining?

Selangor **added 51,626 students** between 2017 and 2025.
Sarawak **lost 43,214 students** during this period.
Large state changes resulted in a small national **decline**.

Which states drove Malaysia's enrolment changes?

Combined government school enrolment change, 2017 - 2025.
Bars show each state's contribution to the national change.



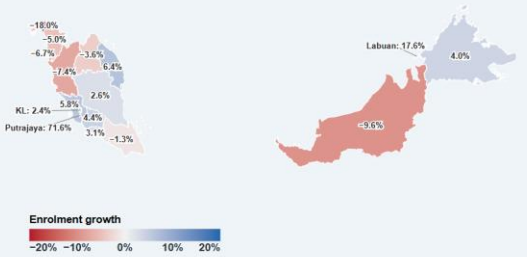
Gains and losses offset one another

Each bar adds another state's change. **Increases** are progressively offset by **decreases**. The final national change is **-4,537 students**.

Which state **grew** fastest?
Which **added** the most students?

Where did school enrolment grow or decline?

Combined government school enrolment change, 2017 - 2025.
Colour scale capped at ±20%.
Hover for exact values.



Where are Malaysia's students concentrated?

Government school enrolment by state, 2025.
Larger circles represent more students.



School stage: Combined

Growth depends on the starting point

Putrajaya's enrolment **rose 71.6%** between 2017 and 2025.
Selangor's enrolment **rose 5.8%** during this period.
A smaller starting population can produce larger percentage changes.

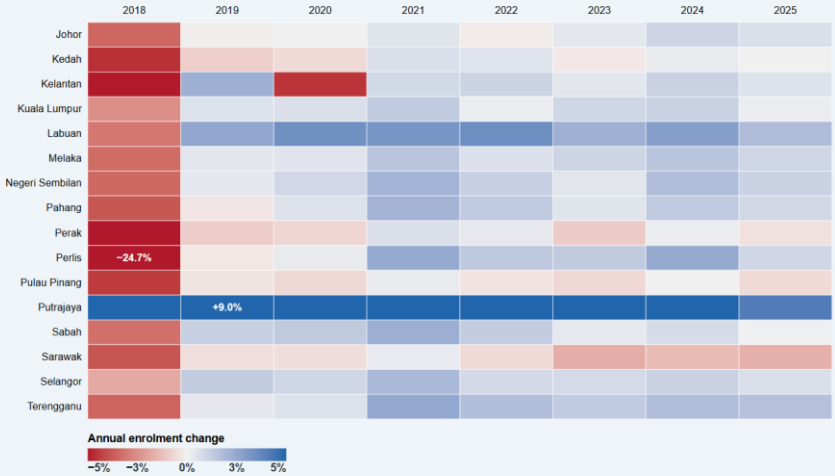
The largest concentration remains in Selangor

Selangor held **19.5%** of national enrolment in 2025.
That represents almost one in five students.
Circle area represents student numbers.

The endpoints show the overall change.
The annual pattern shows when it happened.

When did school enrolment rise or fall?

Annual government school enrolment change, 2018 - 2025.
Primary and **secondary** enrolment are combined.
Each year compares with the previous year.
Colour scale capped at ±5%.



A decline followed by partial recovery

Perlis recorded a **24.7% fall** in 2018.
Its 2025 enrolment remained **18.0% below 2017**.
The colour scale is capped at ±5%.
Larger **declines** share the darkest shade.
Hover for exact values.

WHAT THIS TELLS US

Read student counts, percentage changes and timing together.
Each answers a different question.

THE NEXT GENERATION

03 Today's students. *Tomorrow's school demand.*

Who attends school today?
How is the younger population changing?
Student composition describes the present.
Births and age groups provide future context.

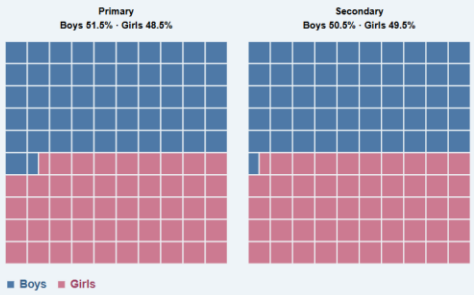
THE QUESTION

What demographic changes accompany enrolment trends?

Boys and **girls** have nearly equal enrolment shares.
Births **fell 18.4%** between 2017 and 2024.
These measures describe different generations.

Who are the students in Malaysia?

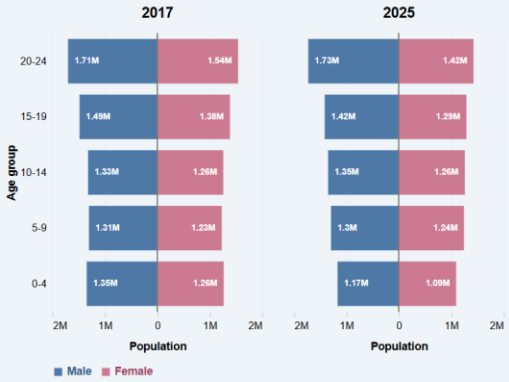
Government **primary** and **secondary** enrolment, 2025.
Each square represents 1%; half represents 0.5%.
Shares are rounded to the nearest 0.5 percentage point.
Blue = **boys**; pink = **girls**.
Hover for exact values.



Selected state: Malaysia

How has the young population in Malaysia changed?

Population aged 0 - 24, by age group and sex.
Compare 2017 with 2025.
Male populations appear left; **female** populations appear right.
Both years use the same population scale.



Selected state: Malaysia

A nearly balanced student population

Girls represented **48.7%** of **primary** enrolment nationally in 2025.
They represented **49.4%** of **secondary** enrolment.
Waffle cells use rounded shares.
Hover for exact values.

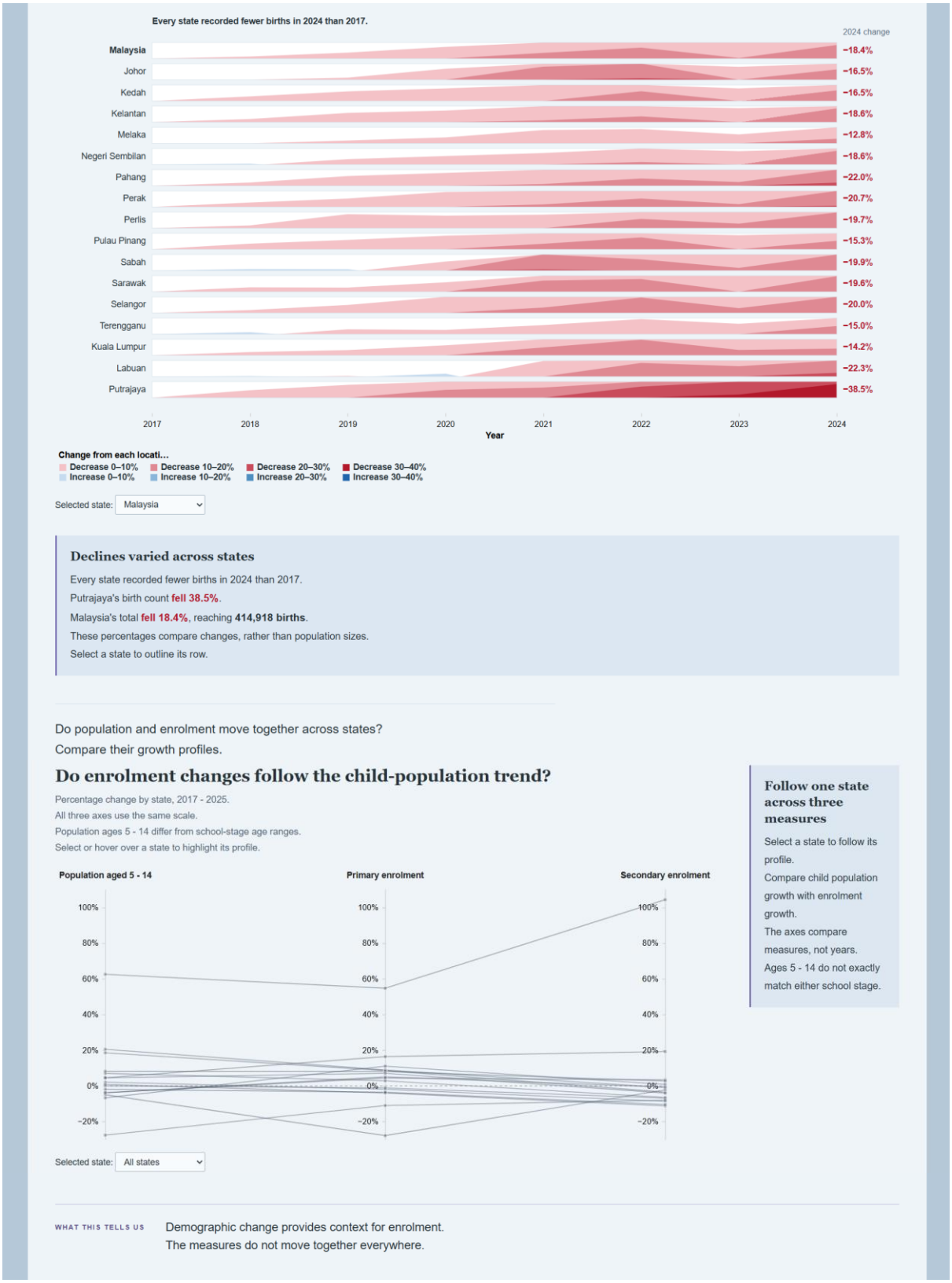
The youngest cohort became smaller

Malaysia's 0 - 4 population numbered about **2.61 million** in 2017.
It **fell** to about **2.26 million** in 2025.
Primary enrolment **increased** over this period.
These measures describe different age groups.

Birth trends provide another view of future school demand.

How have birth trends changed across Malaysian states?

Registered live births compared with each location's 2017 baseline.
Each colour band represents ten percentage points.
Blue indicates **increases**; red indicates **decreases**.
Darker shades show larger changes from 2017.
Right labels show 2024 changes.


Every state recorded fewer births in 2024 than 2017.
2024 change
Malaysia −18.4%
Johor −16.5%
Kedah −16.5%
Kelantan −18.6%
Melaka −12.8%
Negeri Sembilan −18.6%
Pahang −22.0%
Perak −20.7%
Perlis −19.7%
Pulau Pinang −15.3%
Sabah −19.9%
Sarawak −19.6%
Selangor −20.0%
Terengganu −15.0%
Kuala Lumpur −14.2%
Labuan −22.3%
Putrajaya −38.5%
2017 2018 2019 2020 2021 2022 2023 2024
Year
Change from each locati...
Decrease 0–10% Decrease 10–20% Decrease 20–30% Decrease 30–40%
Increase 0–10% Increase 10–20% Increase 20–30% Increase 30–40%
Selected state: Malaysia
Declines varied across states
Every state recorded fewer births in 2024 than 2017.
Putrajaya's birth count fell 38.5%.
Malaysia's total fell 18.4%, reaching 414,918 births.
These percentages compare changes, rather than population sizes.
Select a state to outline its row.
Do population and enrolment move together across states?
Compare their growth profiles.
Do enrolment changes follow the child-population trend?
Percentage change by state, 2017 - 2025.
All three axes use the same scale.
Population ages 5 - 14 differ from school-stage age ranges.
Select or hover over a state to highlight its profile.
Population aged 5 - 14
Primary enrolment
Secondary enrolment
100% 80% 60% 40% 20% 0% −20%
Selected state: All states
Follow one state across three measures
Select a state to follow its profile.
Compare child population growth with enrolment growth.
The axes compare measures, not years.
Ages 5 - 14 do not exactly match either school stage.
WHAT THIS TELLS US
Demographic change provides context for enrolment.
The measures do not move together everywhere.

SCHOOLS & TEACHERS

04 Changing numbers. *Different local needs.*

Student counts describe only part of provision.
Compare schools, teachers and district staffing ratios.
Each measure reveals a different aspect.

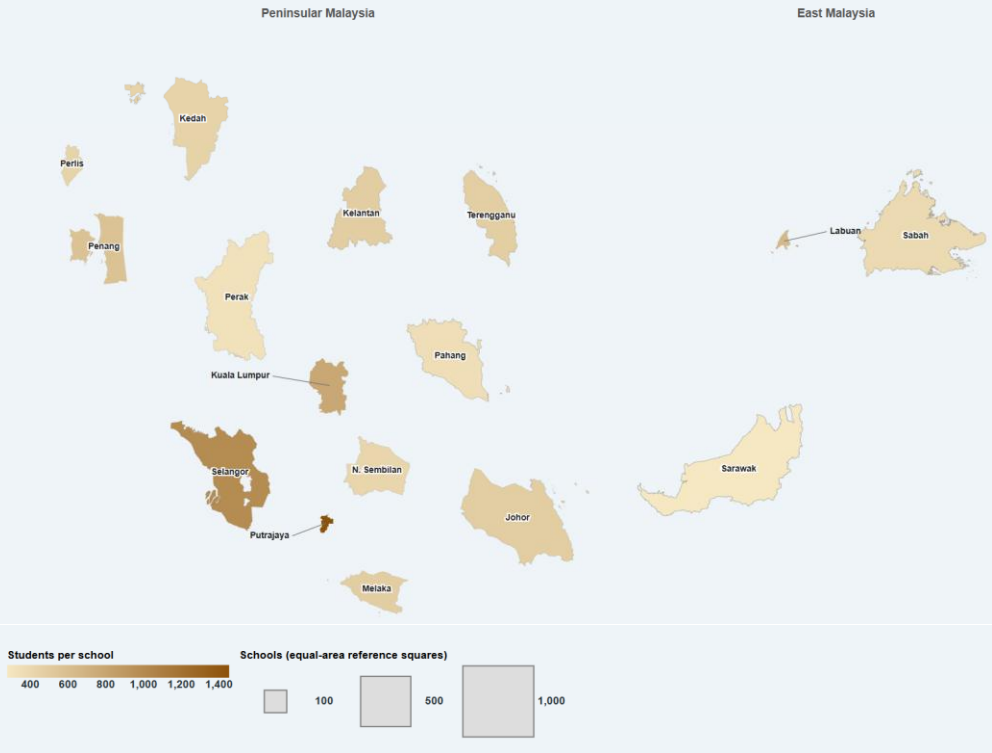
THE QUESTION

How does provision differ across places?

Sarawak had more government schools than Selangor in 2025.
Its average enrolment per school was much lower.
School counts and school sizes reveal different patterns.

How are schools distributed relative to student numbers?

Government **primary** and **secondary** schools by state, 2025.
State shape area represents school count.
Darker brown represents more students per school.
Shapes are resized and separated; positions are approximate.
Grey squares provide school-count area references.



More schools, fewer students per school

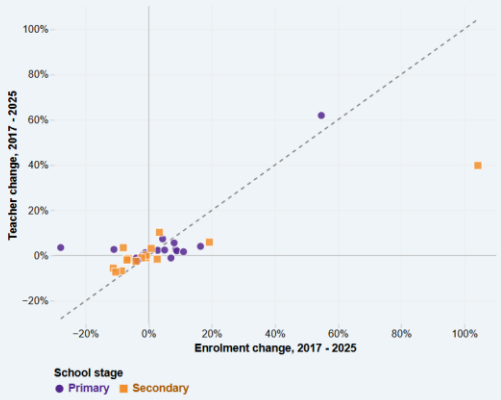
Sarawak had **1,460 schools** in 2025.
It averaged approximately **277 students per school**.
Selangor had **945 schools**.
It averaged approximately **995 students per school**.
These averages do not measure classroom size.

Schools describe one aspect of provision.
Teacher numbers provide another perspective.

Is teacher growth keeping pace with enrolment growth?

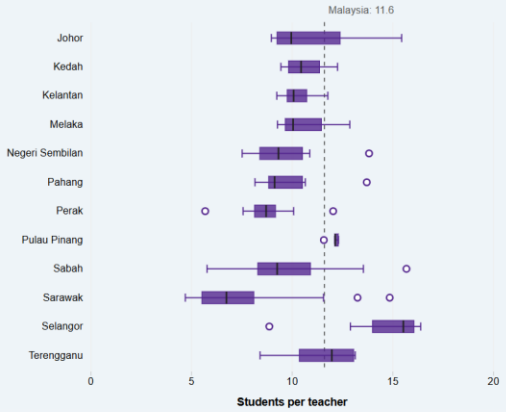
Do state averages hide differences across districts?

Government school percentage change, 2017 - 2025.
Above the diagonal: teacher growth exceeds enrolment growth.
Below the diagonal: enrolment growth exceeds teacher growth.



Selected state: All states

Government school students per teacher, 30 June 2025.
Boxes show the middle half of districts.
Central lines mark district medians.
The dashed line marks Malaysia's overall ratio.



School stage: Primary

Show every district

Compare teacher growth with student growth

National **primary** teacher numbers **rose 2.8%** from 2017 to 2025.
Primary enrolment **rose 2.4%** over this period.
Secondary teacher numbers **fell 0.6%**.
Secondary enrolment **fell 3.2%**.
Use the controls to compare places and school stages.

Look inside each state

Compare **primary** district ratios in 2025.
Selangor's median was **15.5 students per teacher**.
Sarawak's median was **6.8**.
Malaysia's overall ratio was **11.6**.
Boxes show the middle half of districts.
These ratios do not directly measure class size.

WHAT THIS TELLS US

National and state averages provide starting points.
District comparisons reveal differences hidden by these totals.

KEY TAKEAWAYS

A steady total. *Different local stories.*

National stability hides changes across school stages and states.
These comparisons explain why the local picture differs.

01

National stability hides differences.

Primary enrolment **rose 2.4%** from 2017 to 2025.
Secondary enrolment **fell 3.2%** over the same period.
These movements left the national total almost unchanged.

02

State changes balance out.

Selangor **added 51,626 students** between 2017 and 2025.
Sarawak **lost 43,214 students** over the same period.
Compare student counts alongside percentage changes.

03

Local context matters.

Births **fell 18.4%** between 2017 and 2024.
This provides context for future school-age cohorts.
District staffing ratios also vary within states.

School planning needs a local view.
National totals provide a starting point.

SOURCES & SCOPE

Education figures cover Malaysia's government primary and secondary schools. Commentary figures provide national context. They remain national when chart selections change.

1. Education data

Ministry of Education Malaysia. *Student Enrolment, Schools and Teachers by District*, 2017 - 2025. Education records use 1 January for 2017 - 2022. They use 30 June for 2023 - 2025.

2. Population and birth data

Department of Statistics Malaysia (DOSM). *State Population and Annual Live Births*. Population comparisons: 2017 and 2025. Birth records: 2017 - 2024.

A visualisation by Ben Chong Wei Heng | 34860398 | October 2026

Part E: Generative AI Statement

I acknowledged the use of ChatGPT(https://chatgpt.com/) to give suggestions on my project 2 in modified form. I entered the following prompts on 10th of October:

- Gives ideas on what type of chart is best for visualizing.
- Explain if my charts are good enough to satisfy the question.
- Give ideas on how to modify my sentence so that it sounded more professional.
- Check on my webpage whether the title, text, charts, annotations are in the correct layout.